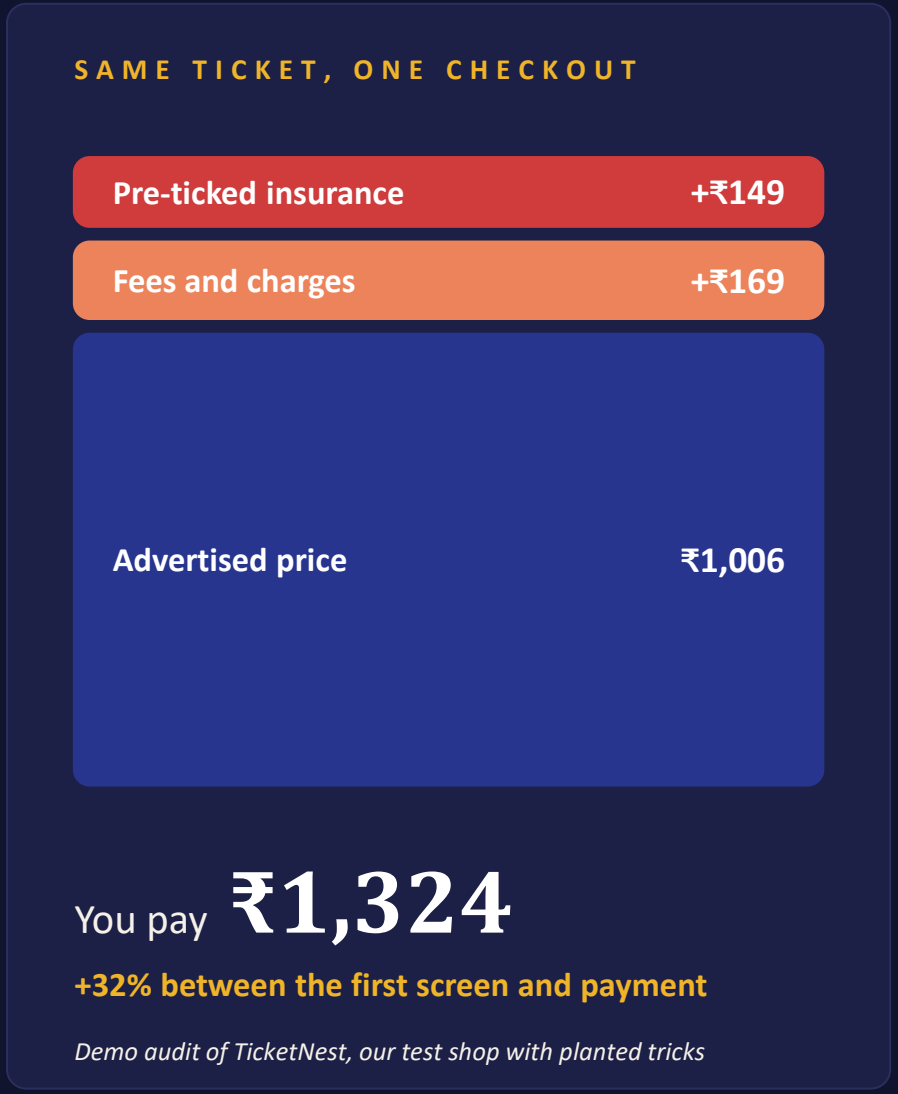
AZINHACK '26 · TEAM RED-HAT

Pardarshi पारदर्शी

AI shopper agents go through checkout like real people, stop before paying, and measure the hidden fees, fake urgency and personalised prices

An active consumer-protection auditor for India's online shoppers

TinyFish | Google ADK | Gemini | scikit-learn | Playwright



Problem Statement

Online prices change between the product page and payment, where a single shopper cannot see it

304M

online buyers in India; 88% lose ₹78–87 a month to hidden charges

₹25–28k cr

lost by Indian online buyers every year to dark patterns and hidden charges

97%

of 290 major Indian platforms still use at least one dark pattern

4.2 lakh

e-commerce complaints a year at the National Consumer Helpline (NCH), 32% of all

Why it persists

Fees appear only at checkout

Drip pricing and pre-ticked add-ons wait on the last screen, after the shopper has committed.

Personalised prices are invisible

One shopper only ever sees their own price, so device or location pricing cannot be noticed.

Compliance is self-declared

The Central Consumer Protection Authority (CCPA) asked 50+ platforms to self-audit in June 2025. No independent check is required.

Sources: 304M, ₹25–28k cr: Datum Intelligence (Jun 2026) · 97%: LocalCircles (Oct 2025) · 4.2 lakh: NCH data FY2024-25 · 50+ platforms: CCPA advisory of 5 Jun 2025 (recipient count as reported in the press).

Proposed Solution

Pardarshi measures what shoppers are actually charged, and turns it into evidence



01

Shop

6 shopper profiles × 3 runs, plus the cancellation flow: 20 runs

02

Stop

Every agent stops before payment, enforced in code on browser lanes

03

Record

Every screen: price lines, pre-ticked boxes, urgency claims, screenshots

04

Measure

ML and statistics turn captures into findings, each with its confidence

05

Act

0–100 trust score, findings report with SHA-256 hashes, NCH complaint draft

Measured, not surveyed. Per-site proof, with screenshots and statistics, ready to attach to a complaint.

Proposed Solution: Seven Checks

Mapped to the Consumer Protection Act 2019 and the CCPA's 2023 dark-pattern guidelines

Drip pricing

First-screen price vs payment total, median of runs

CCPA 2023: drip pricing

Basket sneaking

Paid add-ons that arrive pre-ticked in the basket

CCPA 2023: basket sneaking

Fake urgency

Resetting timers, “only N left”, fake popularity

CCPA 2023: false urgency

Manipulative copy

Confirm-shaming and nagging copy (ML)

CCPA 2023: confirm shaming, nagging

Personalised pricing

Same product, price by device, country, visitor

CPA 2019: unfair trade practice

Cancellation trap

Clicks to cancel against clicks to join

CCPA 2023: subscription trap

Policy vs reality

Written promises the measurements disprove

CPA 2019: misleading advertisement

Every finding carries

its legal category, severity, penalty points, the run it came from and its screenshots.

Proposed Solution: What Is New

Research proved it can be measured; Pardarshi makes it a repeatable, complaint-ready audit for India

Approach	Real charge	Compares profiles	Repeatable + stats	Independent
Consumer surveys (e.g. LocalCircles)	Self-reported	No	No	**Yes**
Studies (Hannak '14, Mathur '19)	**Yes**	**Yes**	One-off studies	**Yes**
Detectors and extensions	One page, one shopper	No	No	**Yes**
Manual investigation	**Yes**	Rarely	No	**Yes**
Platform self-audit (CCPA, 2025)	Not disclosed	Not disclosed	Not disclosed	No
Pardarshi	**Yes, at checkout**	**Yes, 6 profiles**	**Per site, on demand**	**Yes**

Evidence, not opinion

Screenshots, price lines and SHA-256 hashes behind every finding.

Several shopper profiles

Personalised pricing only shows when you compare profiles.

Ready to act on

Findings map to Indian law and become an NCH complaint.

Target Users and Use Case

Built for people who need proof, not opinions

PRIMARY

Consumer-rights investigators and journalists

Today: burner accounts, manual screenshots, a single shopper's view.

With Pardarshi: 20 shopper runs (35 seconds on our demo shop), statistics, a case file.

SECONDARY

Platform compliance teams

Today: the CCPA's 2025 advisory asks for a self-audit and a self-declaration.

With Pardarshi: an independent, repeatable check of their own site, with consent.

BENEFICIARY

Shoppers and regulators

Today: vague complaints (“they overcharged me”) that are hard to act on.

With Pardarshi: complaints that name the category, the figures and the exhibits.

Demo use case Our demo audit of TicketNest (a test shop with planted tricks): ₹1,006 became ₹1,324 at payment (+32%), iPhone shoppers were quoted 9.6% more, and cancelling took 8 clicks against 1 to join. Out came a findings report and an NCH complaint.

Role of Machine Learning

A trained classifier, statistics and rules turn captures into findings, measured on unseen data

Dark-pattern copy classifier (trained)

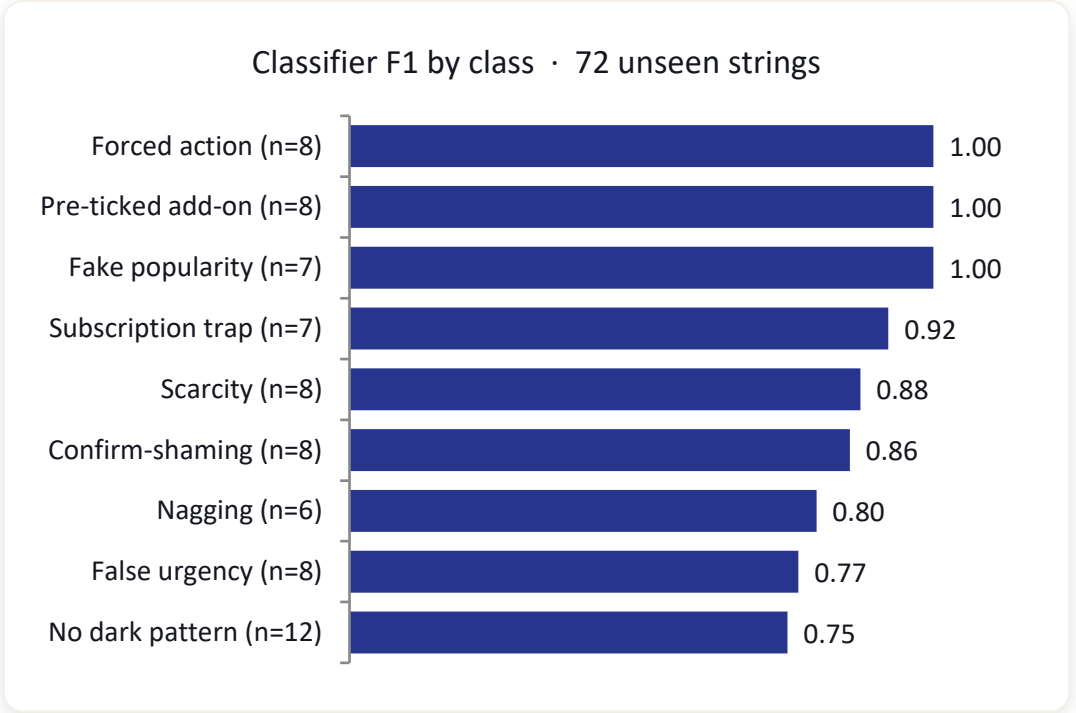
TF-IDF word and character n-grams into logistic regression; 9 classes, 725 labelled UI strings (seed set plus augmentation). Tested on 72 human-written strings never seen in training.

Personalised-pricing statistics

Regression on log price by device, country and visitor, with 95% confidence intervals and Benjamini–Hochberg correction. Runs go in random interleaved rounds, so drift over time cannot pose as personalisation.

Price extraction and trust score (rules)

Page text into labelled lines: item, fee, tax, add-on, total. Score = 100 × e^(−points / 70), each point traced to a finding.



0.89
macro-F1 on unseen strings

1.00
precision, dark vs none (recall 0.87)

12/12
planted tricks found (demo shop)

Per-class n is small (6–12), so these are early numbers. 12/12 is on our own shop with planted tricks; real-site validation is next.

Role of AI

Gemini and Google ADK do the reasoning; guards keep the agents from ever paying

Shopper agent (Gemini)

Reads each page and picks one action at a time towards checkout, on sites nobody scripted. TinyFish's Agent API does this in the cloud.

Guards that stop payment

Browser lanes: code blocks Pay, Confirm order, Subscribe and never types card, OTP or personal data. Agent API: no account, address or card to pay with.

Policy vs reality (Gemini)

Reads the site's terms and refund pages and flags written promises that the measured runs disprove.

Google ADK orchestrator

A tool-using agent that runs audits, finds peer sites (TinyFish Search), reads policies (Fetch) and drafts complaints.

Voice and text assistant

Exact answers from the audit's own numbers; open questions go to Gemini with the whole audit as context.

Complaint writer

Drafts a National Consumer Helpline complaint that cites only measured findings, with exhibits.

Proposed Technical Approach

One pipeline, two browser providers, evidence captured at every step

Console

FastAPI, live agent view, voice and text assistant

Orchestrator

Google ADK + pipeline: 6 profiles × 3 runs in random rounds, one product

Browsers

TinyFish Browser and Agent API, or Playwright (open source)

Capture

Every screen: prices, copy, checkboxes, screenshots

Analysis

Classifier, price statistics, Gemini policy check

Outputs

Score, findings report, SHA-256 hashes, NCH complaint

Stack

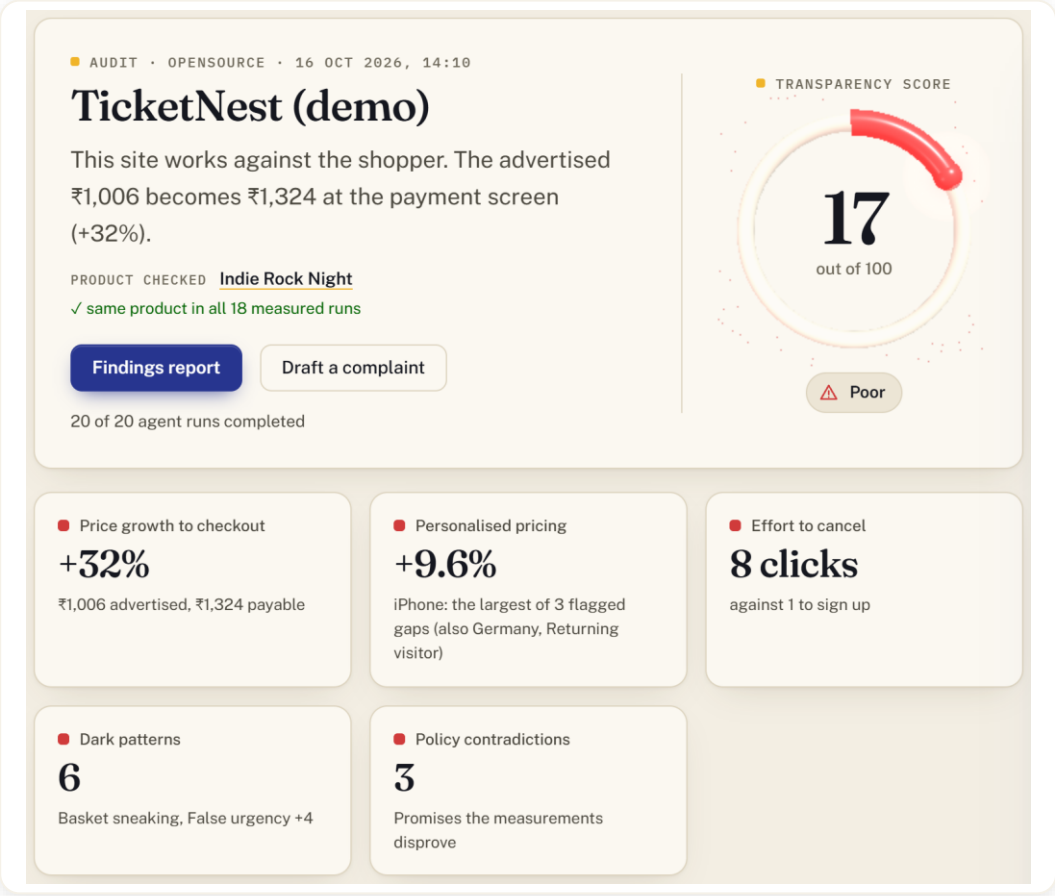
Python · FastAPI · TinyFish SDK · Playwright · Google ADK · Gemini · scikit-learn · SciPy · SQLite

Safety by design

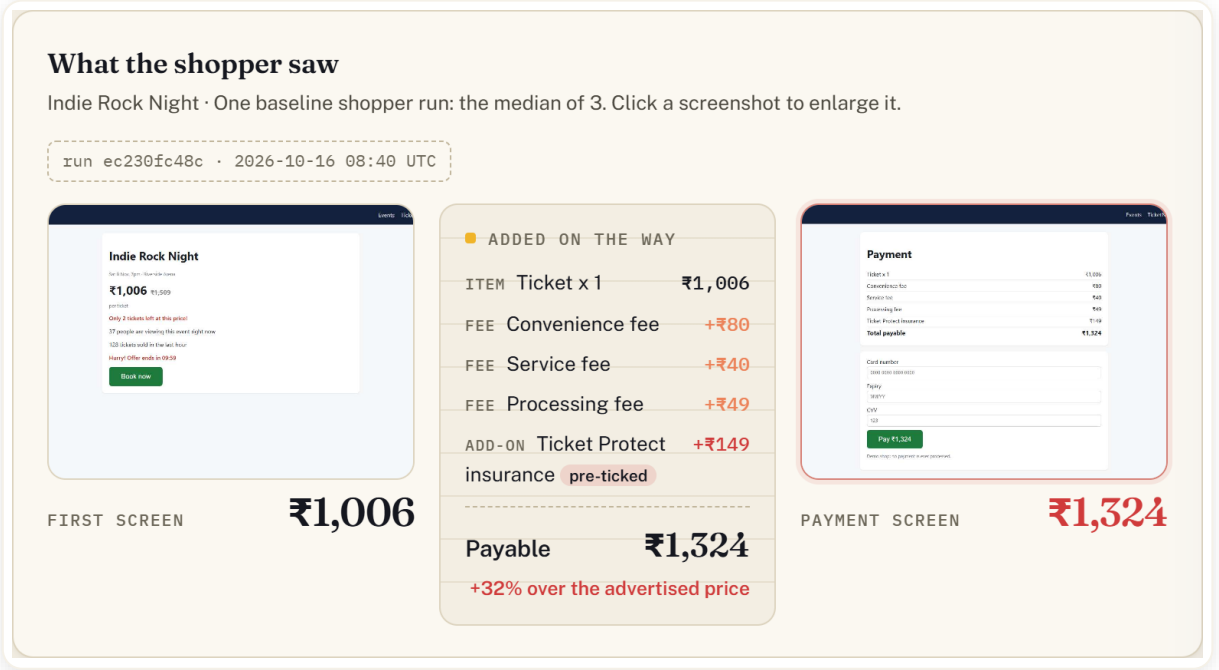
- Allowlisted sites only
- Stops before payment
- Pay-button guard in code
- One shopper at a time
- Blocks reported, not bypassed
- Private, per-user audits

Proposed Technical Approach: Working Product

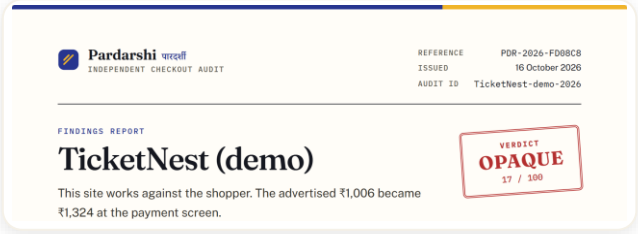
One demo audit, three real screens: no mock-ups



Console: score, verdict, same product in all 18 runs, key measurements



Evidence: one real run, first screen vs payment, every line added



Findings report
Reference, verdict stamp, exhibits and an integrity section with SHA-256 hashes

Proposed TinyFish Integration

All four TinyFish APIs integrated, each with a specific job, and verified live on 16 Oct 2026

Browser API

per shopper run

A cloud browser per shopper over the Chrome DevTools Protocol (CDP), with a country proxy (US, GB, CA, DE, FR, JP, AU). Our flows set device, locale and timezone, take screenshots and reload to expose fake countdowns.

Agent API

unscripted sites

A goal (reach checkout, never pay) plus a typed output_schema: advertised price, each price line with its pre-ticked flag, verbatim urgency and opt-out copy, the product and where it stopped.

Fetch API

policy pages

Terms, refund and cancellation pages as clean markdown, up to 10 URLs per call, feeding the Gemini policy-vs-reality check.

Search API

peer sites

Finds comparable services on the live web for the Google ADK agent's peer-site tool, so a site can be compared with its peers.

Live proof, 16 Oct 2026: Browser session tf-2c5a76ec… exit IP country DE · Agent run COMPLETED in 5 steps, JSON {"heading": "Example Domain"}
Cost per 20-run audit: ≈ $0.04 Browser API (scripted sites) · ≈ $5 Agent API (any site) · Search, Fetch $0 (TinyFish rates, 16 Oct 2026)

Proposed TinyFish Integration: One Run

One shopper run through TinyFish, step by step

1. **Pick a shopper**
   iPhone · Germany · new visitor
2. **Open a session**
   Browser API with a DE proxy (verified live), or the Agent API
3. **Become that shopper**
   CDP: device metrics, user agent, locale, timezone
4. **Walk the checkout**
   Pinned product → basket → checkout, every screen captured
5. **Stop safely**
   At payment, a login wall or a CAPTCHA (reported, never bypassed)
6. **Hand back evidence**
   Typed JSON + screenshots → RunRecord → ML and statistics

Agent API · illustrative output, demo-audit values


```
{
  "product_name": "Indie Rock Night",
  "advertised_price": 1006,
  "final_payable_total": 1324,
  "stopped_at": "payment",
  "steps": [{
    "name": "cart",
    "price_lines": [
      {"label": "Convenience fee", "amount": 80},
      {"label": "Ticket Protect", "amount": 149,
       "preselected": true}]
  }]
}
```


Why TinyFish

- **Real geography:** a German exit IP, verified live; India exits are on our roadmap.
- **Unscripted sites:** the Agent API walks checkouts nobody scripted.
- **Managed browsers:** we spend our time on measurement, not infrastructure.

Expected Impact

From a shopper's suspicion to evidence a regulator can act on

One shopper's view → **20 runs in 35 s**

measured on our demo shop; real sites run one shopper at a time

Suspicion → **Statistics**

price gaps with 95% confidence intervals and a noise floor

Complaint → **Case file**

an NCH-ready letter with exhibits and SHA-256 hashes

Self-declaration → **Independent check**

a repeatable re-check that a regulator can ask for

Pilot goal **2 consumer NGOs · 50 Indian sites in 3 months**
Budget ≈ ₹22,000 on the TinyFish Agent API (≈ ₹440 per audit)

+32%

price growth from the first screen to payment

+9.6%

quoted to iPhone shoppers (95% CI 8.4–10.9%)

8 vs 1

clicks to cancel against clicks to join

17/100

trust score: 12 findings, 20/20 runs

All four figures: our demo audit of TicketNest, a test shop with planted tricks.

Feasibility

Built, tested and running today, with an honest list of risks

Built and working

- End to end: 20 of 20 runs in 35 s (demo shop)
- All four TinyFish APIs verified live (16 Oct 2026)
- Real site (Demoblaze): reached payment, no false alarms
- 92 automated tests: ML, flows, guards, auth, TinyFish
- Classifier on unseen strings: macro-F1 0.89
- Private accounts, live console, printable report

Risk	Mitigation
Bot protection	Reported with evidence, never bypassed; standard (not stealth) browsers; own-account session
Login before payment	Measure up to the furthest price screen; audit your own account
False positives	Noise floor, Benjamini–Hochberg, 2% minimum gap, random interleaved run order; a person files
Evidence in court	SHA-256 hashes reveal tampering; next: trusted timestamps and a Section 63 (BSA 2023) certificate
Legal and ethics	Allowlisted sites, observe only, never pays, one shopper at a time
Cost	₹0 open-source; ≈ ₹3.5 per audit on the Browser API (scripted), ≈ ₹440 on the Agent API

Future Scope

From a hackathon tool to public consumer-protection infrastructure

NEXT 3 MONTHS

- Validate on real Indian sites with consumer NGOs
- India exit nodes for geo-pricing tests
- Hindi and Indic-language copy classifier

6 MONTHS

- Shopper capture extension: real people, no bot
- Continuous monitoring with alerts
- Regulator dashboard across sites

12 MONTHS

- Platform allowlisting for independent audits
- Filing into NCH's INGRAM complaint portal
- Apps and quick-commerce audits

Sustainability free for shoppers and consumer NGOs; paid, consented recurring audits for platforms that want to prove compliance.

Measured, not surveyed

Pardarshi sends AI shoppers through checkout, stops before paying, and hands consumers and regulators the evidence: the hidden fees, the fake urgency and the personalised prices.

4
TinyFish APIs, verified live

7
checks mapped to Indian law

0.89
classifier macro-F1, unseen text

92
automated tests

Thank you पारदर्शी